Linkit Kim/SMD

linkit-v4

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Legend

General comment such as function title, configuration, ...

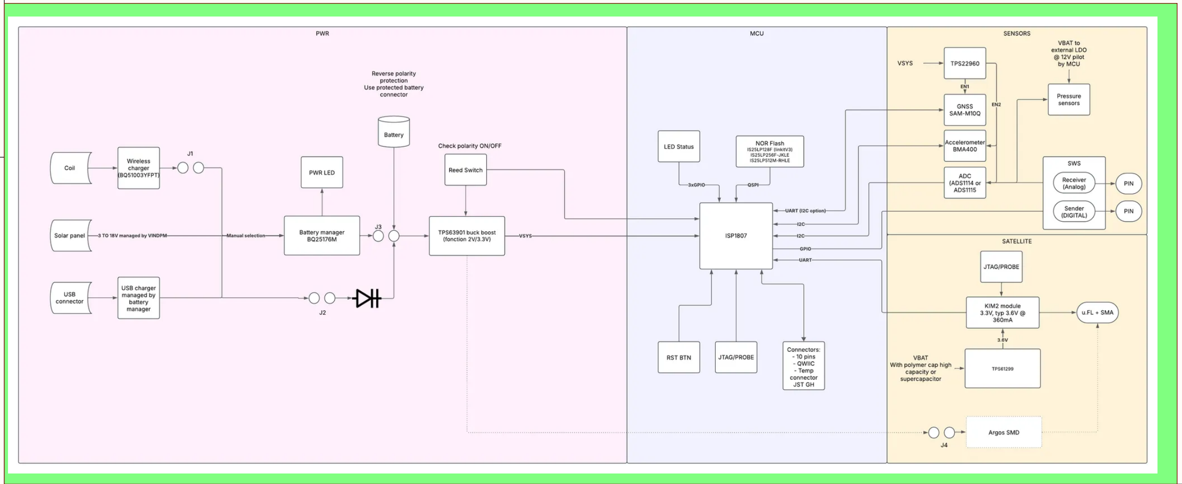
Text to be added to silkscreen.

Warning text.

Notes to generate the board layout.

License

U_linkit-v4
linkit-v4.SchDoc



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Size: A4	Reference: linkit-v4	Revision: 1 0	
Date: 14/11/2025	Time: 13:03:49	Sheet 1 of 13	
File: C:\Users\fourn\Altium\AD25\Projects\Arribada\linkit-v4-hw\Project_Overview.SchDoc			

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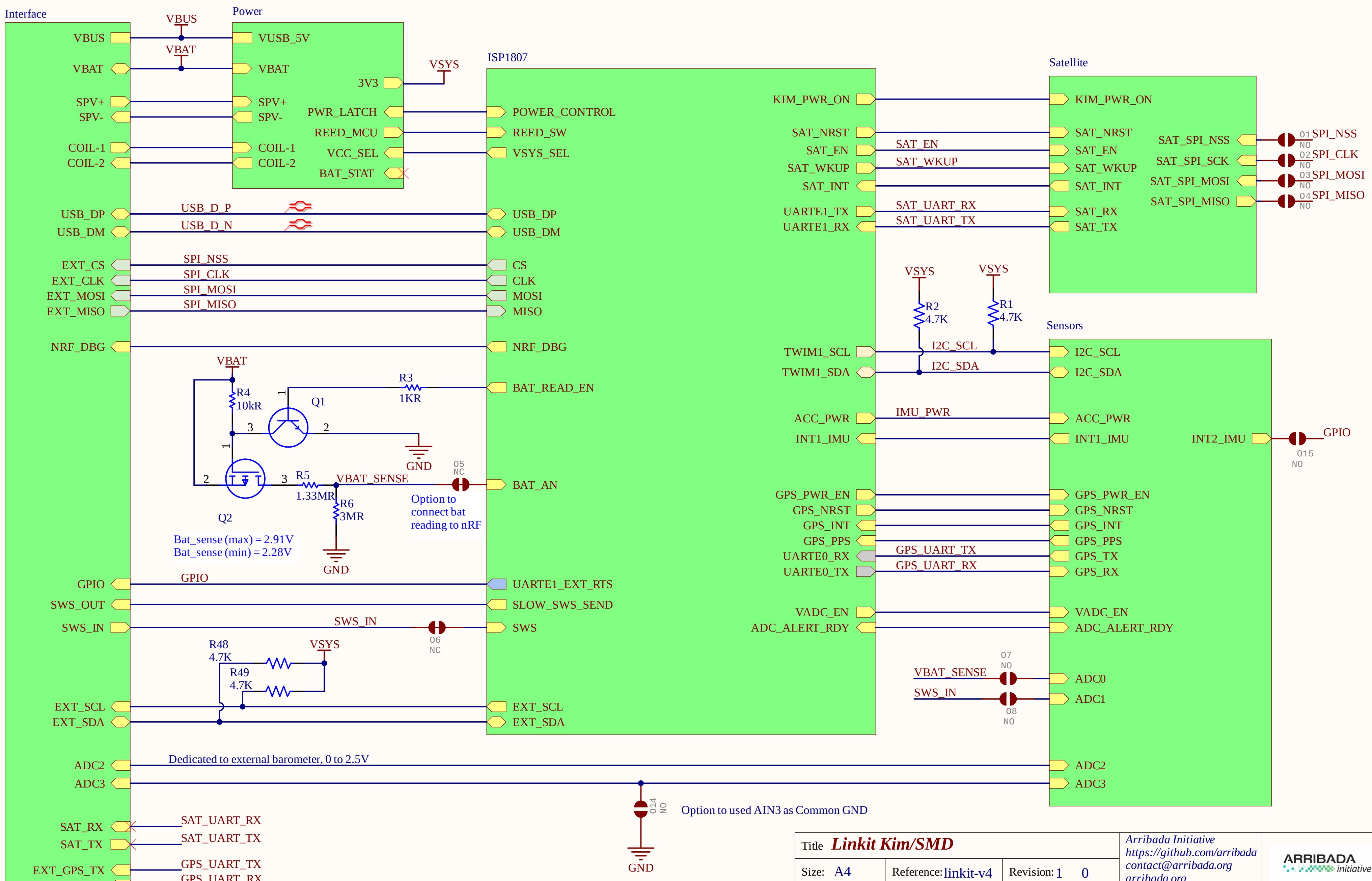

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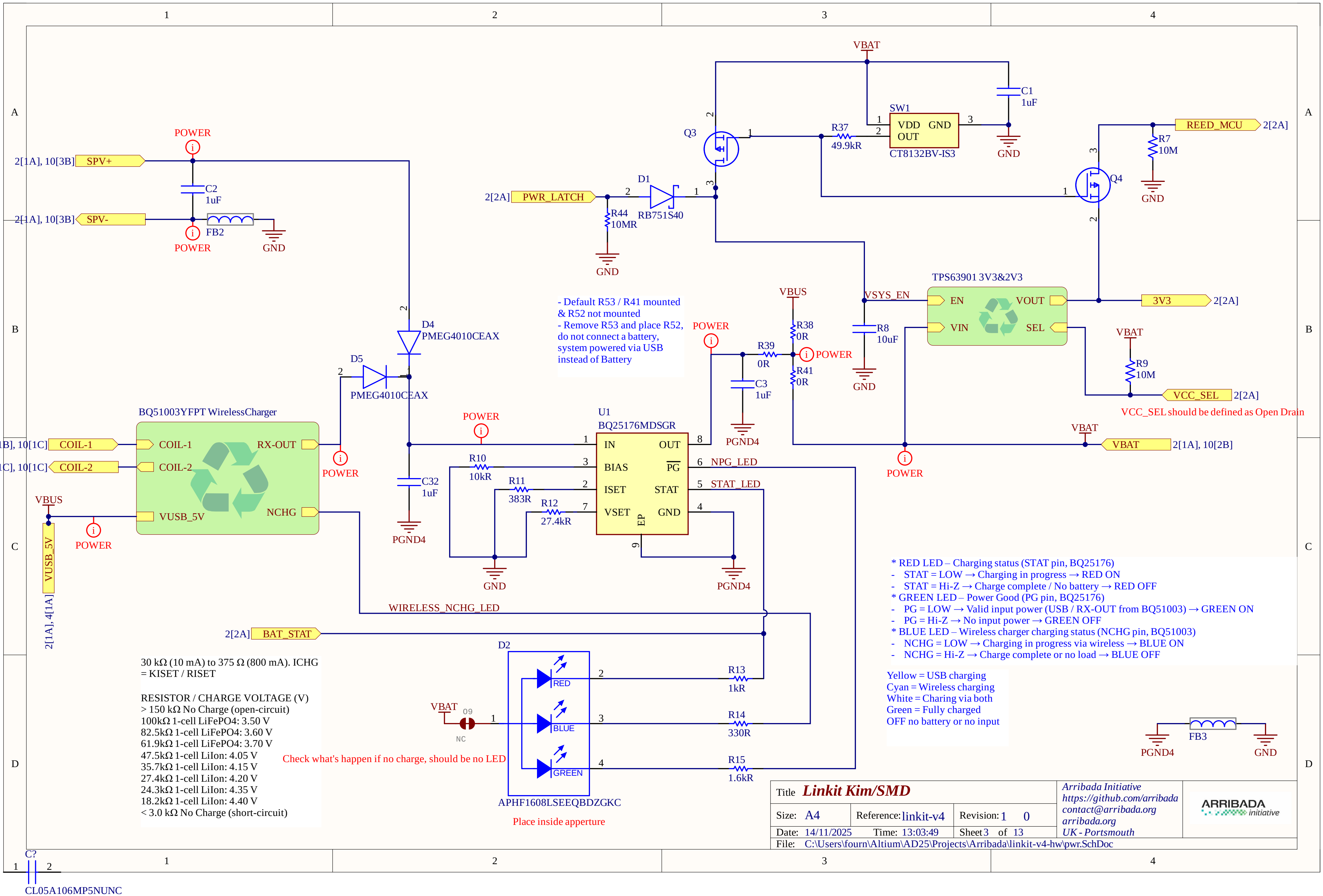
VBAT
C1 1uF
SW1
VDD GND OUT
CT8132BV-IS3
GND
R37 49.9kR
Q3
REED_MCU 2[2A]
R7 10M
Q4
GND
POWER
2[1A], 10[3B] SPV+
C2 1uF
2[1A], 10[3B] SPV-
FB2
POWER
GND
D1
2[2A] PWR_LATCH
R44 10MR
RB751S40
GND
VSYS_EN
TPS63901 3V3&2V3
EN VOUT VIN SEL
3V3 2[2A]
R8 10uF
GND
VBUS
R38 0R
R39 0R
R41 0R
POWER
C3 1uF
PGND4
VBAT
R9 10M
VCC_SEL 2[2A]
VCC_SEL should be defined as Open Drain
- Default R53 / R41 mounted & R52 not mounted
- Remove R53 and place R52, do not connect a battery, system powered via USB instead of Battery
D4 PMEG4010CEAX
D5 PMEG4010CEAX
BQ51003YFPT WirelessCharger
COIL-1 RX-OUT
COIL-2
VUSB_5V NCHG
[1B], 10[1C] COIL-1
[1C], 10[1C] COIL-2
VBUS
POWER
VUSB_5V
2[1A], 4[1A]
C32 1uF
PGND4
U1
BQ25176MDSGR
IN OUT
BIAS PG
ISET STAT
VSET GND
EP
R10 10kR
R11 383R
R12 27.4kR
GND
PGND4
NPG_LED
STAT_LED
VBAT 2[1A], 10[2B]
POWER
WIRELESS_NCHG_LED
2[2A] BAT_STAT
D2
RED
BLUE
GREEN
APHF1608LSEEQBDZGKC
Place inside apperture
R13 1kR
R14 330R
R15 1.6kR
VBAT
NC
Check what's happen if no charge, should be no LED
30 kΩ (10 mA) to 375 Ω (800 mA). ICHG = KISET / RISET
RESISTOR / CHARGE VOLTAGE (V)
> 150 kΩ No Charge (open-circuit)
100kΩ 1-cell LiFePO4: 3.50 V
82.5kΩ 1-cell LiFePO4: 3.60 V
61.9kΩ 1-cell LiFePO4: 3.70 V
47.5kΩ 1-cell LiIon: 4.05 V
35.7kΩ 1-cell LiIon: 4.15 V
27.4kΩ 1-cell LiIon: 4.20 V
24.3kΩ 1-cell LiIon: 4.35 V
18.2kΩ 1-cell LiIon: 4.40 V
< 3.0 kΩ No Charge (short-circuit)
* RED LED – Charging status (STAT pin, BQ25176)
- STAT = LOW → Charging in progress → RED ON
- STAT = Hi-Z → Charge complete / No battery → RED OFF
* GREEN LED – Power Good (PG pin, BQ25176)
- PG = LOW → Valid input power (USB / RX-OUT from BQ51003) → GREEN ON
- PG = Hi-Z → No input power → GREEN OFF
* BLUE LED – Wireless charger charging status (NCHG pin, BQ51003)
- NCHG = LOW → Charging in progress via wireless → BLUE ON
- NCHG = Hi-Z → Charge complete or no load → BLUE OFF
Yellow = USB charging
Cyan = Wireless charging
White = Charing via both
Green = Fully charged
OFF no battery or no input
FB3
PGND4
GND
C?
CL05A106MP5NUNC
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Size: A4
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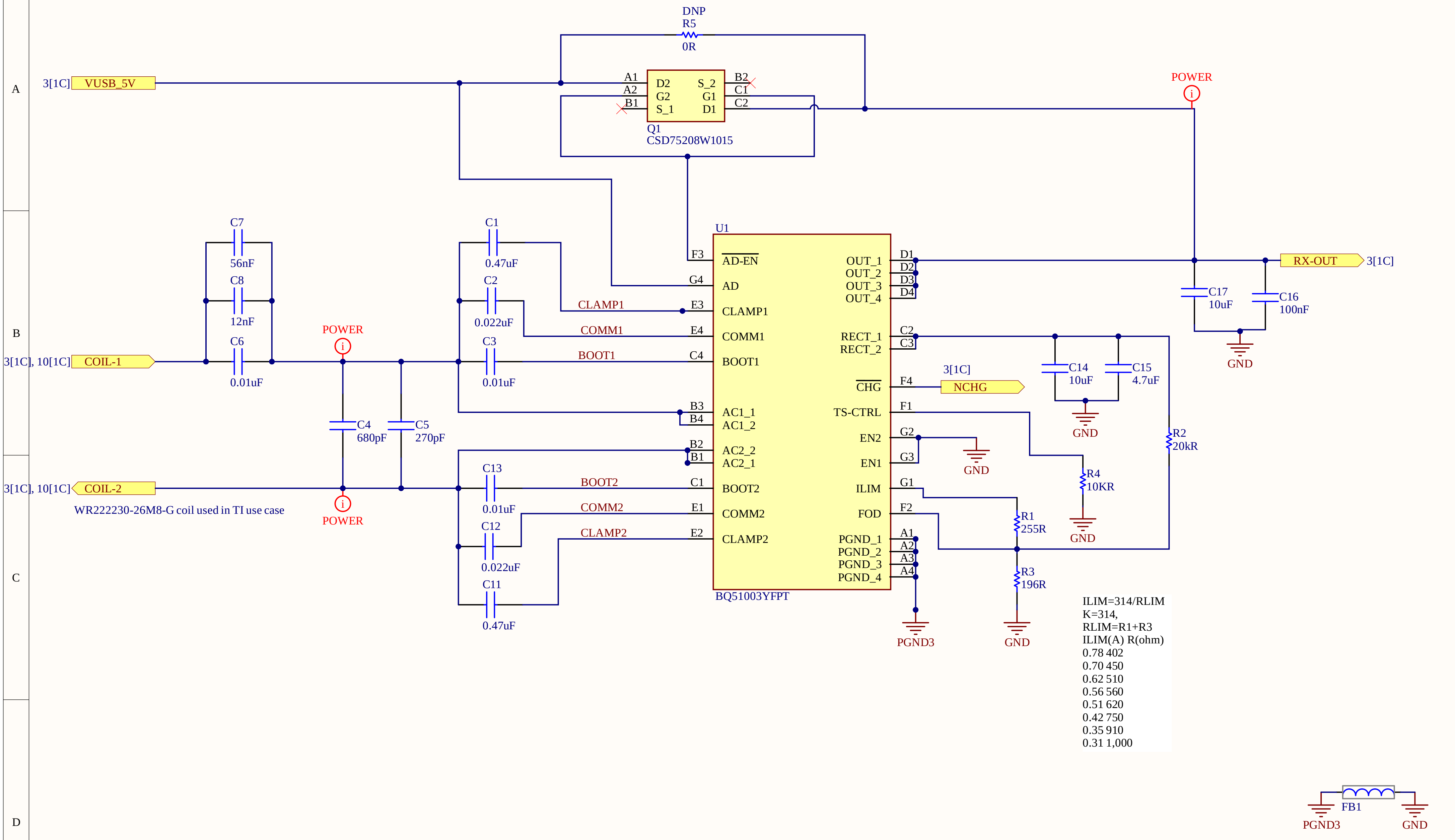

ILIM=314/RLIM
K=314,
RLIM=R1+R3

ILIM(A)	R(ohm)
0.78	402
0.70	450
0.62	510
0.56	560
0.51	620
0.42	750
0.35	910
0.31	1,000

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File: C:\Users\fourn\Altium\AD25\Projects\Arribada\hardware-lib\DeviceSheets\Power\BQ51003YFPT_WirelessCharger_2.5W.S

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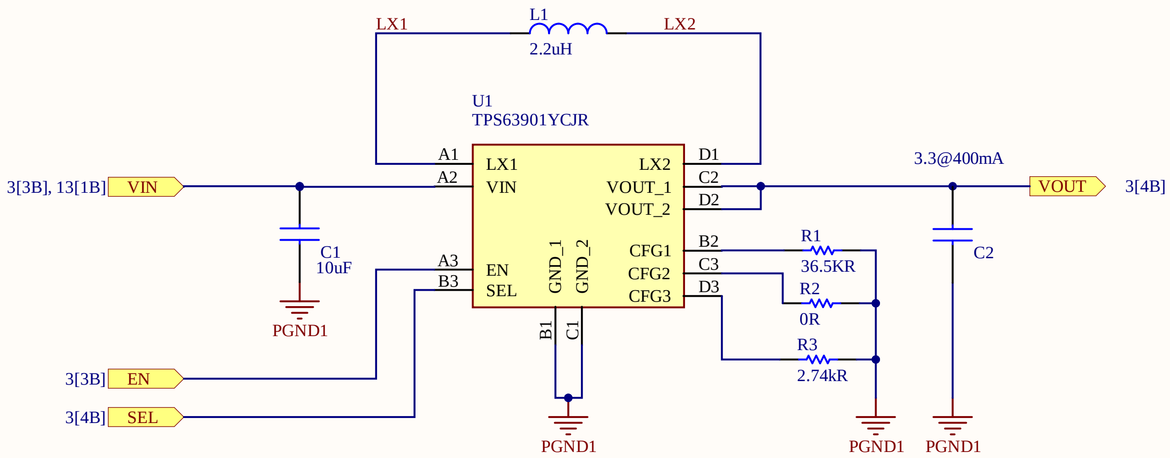
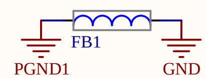


Output Voltage – $V_{O(2)}$ (SEL = HIGH)		Input Current Limit							
		UNLIMITED	100 mA	50 mA	25 mA	10 mA	5 mA	2.5 mA	1 mA
1.8 V	R_{CFG1}	0 Ω							
	R_{CFG2}	0 Ω	511 Ω	1.15 kΩ	1.87 kΩ	2.74 kΩ	3.83 kΩ	5.11 kΩ	6.49 kΩ
1.9 V	R_{CFG1}	511 Ω							
	R_{CFG2}	0 Ω	511 Ω	1.15 kΩ	1.87 kΩ	2.74 kΩ	3.83 kΩ	5.11 kΩ	6.49 kΩ
2.0 V	R_{CFG1}	1.15 kΩ							
	R_{CFG2}	0 Ω	511 Ω	1.15 kΩ	1.87 kΩ	2.74 kΩ	3.83 kΩ	5.11 kΩ	6.49 kΩ
2.1 V	R_{CFG1}	1.87 kΩ							
	R_{CFG2}	0 Ω	511 Ω	1.15 kΩ	1.87 kΩ	2.74 kΩ	3.83 kΩ	5.11 kΩ	6.49 kΩ
2.2 V	R_{CFG1}	2.74 kΩ							
	R_{CFG2}	0 Ω	511 Ω	1.15 kΩ	1.87 kΩ	2.74 kΩ	3.83 kΩ	5.11 kΩ	6.49 kΩ
2.3 V	R_{CFG1}	3.83 kΩ							
	R_{CFG2}	0 Ω	511 Ω	1.15 kΩ	1.87 kΩ	2.74 kΩ	3.83 kΩ	5.11 kΩ	6.49 kΩ
2.4 V	R_{CFG1}	5.11 kΩ							
	R_{CFG2}	0 Ω	511 Ω	1.15 kΩ	1.87 kΩ	2.74 kΩ	3.83 kΩ	5.11 kΩ	6.49 kΩ
2.5 V	R_{CFG1}	6.49 kΩ							
	R_{CFG2}	0 Ω	511 Ω	1.15 kΩ	1.87 kΩ	2.74 kΩ	3.83 kΩ	5.11 kΩ	6.49 kΩ
2.6 V	R_{CFG1}	8.25 kΩ							
	R_{CFG2}	0 Ω	511 Ω	1.15 kΩ	1.87 kΩ	2.74 kΩ	3.83 kΩ	5.11 kΩ	6.49 kΩ
2.7 V	R_{CFG1}	10.5 kΩ							
	R_{CFG2}	0 Ω	511 Ω	1.15 kΩ	1.87 kΩ	2.74 kΩ	3.83 kΩ	5.11 kΩ	6.49 kΩ
2.8 V	R_{CFG1}	13.3 kΩ							
	R_{CFG2}	0 Ω	511 Ω	1.15 kΩ	1.87 kΩ	2.74 kΩ	3.83 kΩ	5.11 kΩ	6.49 kΩ
2.9 V	R_{CFG1}	16.2 kΩ							
	R_{CFG2}	0 Ω	511 Ω	1.15 kΩ	1.87 kΩ	2.74 kΩ	3.83 kΩ	5.11 kΩ	6.49 kΩ
3.0 V	R_{CFG1}	20.5 kΩ							
	R_{CFG2}	0 Ω	511 Ω	1.15 kΩ	1.87 kΩ	2.74 kΩ	3.83 kΩ	5.11 kΩ	6.49 kΩ
3.1 V	R_{CFG1}	24.9 kΩ							
	R_{CFG2}	0 Ω	511 Ω	1.15 kΩ	1.87 kΩ	2.74 kΩ	3.83 kΩ	5.11 kΩ	6.49 kΩ
3.2 V	R_{CFG1}	30.1 kΩ							
	R_{CFG2}	0 Ω	511 Ω	1.15 kΩ	1.87 kΩ	2.74 kΩ	3.83 kΩ	5.11 kΩ	6.49 kΩ
3.3 V	R_{CFG1}	36.5 kΩ							
	R_{CFG2}	0 Ω	511 Ω	1.15 kΩ	1.87 kΩ	2.74 kΩ	3.83 kΩ	5.11 kΩ	6.49 kΩ
3.4 V	R_{CFG1}	0 Ω							
	R_{CFG2}	8.25 kΩ	10.5 kΩ	13.3 kΩ	16.2 kΩ	20.5 kΩ	24.9 kΩ	30.1 kΩ	36.5 kΩ
3.5 V	R_{CFG1}	511 Ω							
	R_{CFG2}	8.25 kΩ	10.5 kΩ	13.3 kΩ	16.2 kΩ	20.5 kΩ	24.9 kΩ	30.1 kΩ	36.5 kΩ
3.6 V	R_{CFG1}	1.15 kΩ							
	R_{CFG2}	8.25 kΩ	10.5 kΩ	13.3 kΩ	16.2 kΩ	20.5 kΩ	24.9 kΩ	30.1 kΩ	36.5 kΩ
3.7 V	R_{CFG1}	1.87 kΩ							
	R_{CFG2}	8.25 kΩ	10.5 kΩ	13.3 kΩ	16.2 kΩ	20.5 kΩ	24.9 kΩ	30.1 kΩ	36.5 kΩ
3.8 V	R_{CFG1}	2.74 kΩ							
	R_{CFG2}	8.25 kΩ	10.5 kΩ	13.3 kΩ	16.2 kΩ	20.5 kΩ	24.9 kΩ	30.1 kΩ	36.5 kΩ
3.9 V	R_{CFG1}	3.83 kΩ							
	R_{CFG2}	8.25 kΩ	10.5 kΩ	13.3 kΩ	16.2 kΩ	20.5 kΩ	24.9 kΩ	30.1 kΩ	36.5 kΩ
4.0 V	R_{CFG1}	5.11 kΩ							
	R_{CFG2}	8.25 kΩ	10.5 kΩ	13.3 kΩ	16.2 kΩ	20.5 kΩ	24.9 kΩ	30.1 kΩ	36.5 kΩ
4.1 V	R_{CFG1}	6.49 kΩ							
	R_{CFG2}	8.25 kΩ	10.5 kΩ	13.3 kΩ	16.2 kΩ	20.5 kΩ	24.9 kΩ	30.1 kΩ	36.5 kΩ
4.2 V	R_{CFG1}	8.25 kΩ							
	R_{CFG2}	8.25 kΩ	10.5 kΩ	13.3 kΩ	16.2 kΩ	20.5 kΩ	24.9 kΩ	30.1 kΩ	36.5 kΩ
4.3 V	R_{CFG1}	10.5 kΩ							
	R_{CFG2}	8.25 kΩ	10.5 kΩ	13.3 kΩ	16.2 kΩ	20.5 kΩ	24.9 kΩ	30.1 kΩ	36.5 kΩ
4.4 V	R_{CFG1}	13.3 kΩ							
	R_{CFG2}	8.25 kΩ	10.5 kΩ	13.3 kΩ	16.2 kΩ	20.5 kΩ	24.9 kΩ	30.1 kΩ	36.5 kΩ
4.5 V	R_{CFG1}	16.2 kΩ							
	R_{CFG2}	8.25 kΩ	10.5 kΩ	13.3 kΩ	16.2 kΩ	20.5 kΩ	24.9 kΩ	30.1 kΩ	36.5 kΩ
4.6 V	R_{CFG1}	20.5 kΩ							
	R_{CFG2}	8.25 kΩ	10.5 kΩ	13.3 kΩ	16.2 kΩ	20.5 kΩ	24.9 kΩ	30.1 kΩ	36.5 kΩ
4.7 V	R_{CFG1}	24.9 kΩ							
	R_{CFG2}	8.25 kΩ	10.5 kΩ	13.3 kΩ	16.2 kΩ	20.5 kΩ	24.9 kΩ	30.1 kΩ	36.5 kΩ
4.8 V	R_{CFG1}	30.1 kΩ							
	R_{CFG2}	8.25 kΩ	10.5 kΩ	13.3 kΩ	16.2 kΩ	20.5 kΩ	24.9 kΩ	30.1 kΩ	36.5 kΩ
5.0 V	R_{CFG1}	36.5 kΩ							
	R_{CFG2}	8.25 kΩ	10.5 kΩ	13.3 kΩ	16.2 kΩ	20.5 kΩ	24.9 kΩ	30.1 kΩ	36.5 kΩ

Table 6-3. Output Voltage (SEL Pin = Low) Settings

Output Voltage – $V_{O(1)}$ (SEL = LOW)	R_{CFG3}
1.8 V	0 Ω
2.0 V	511 Ω
2.1 V	1.15 kΩ
2.2 V	1.87 kΩ
2.3 V	2.74 kΩ
2.4 V	3.83 kΩ
2.5 V	5.11 kΩ
2.6 V	6.49 kΩ
2.7 V	8.25 kΩ
2.8 V	10.5 kΩ
3.0 V	13.3 kΩ
3.3 V	16.2 kΩ
3.6 V	20.5 kΩ
4.0 V	24.9 kΩ
4.5 V	30.1 kΩ
5.0 V	36.5 kΩ



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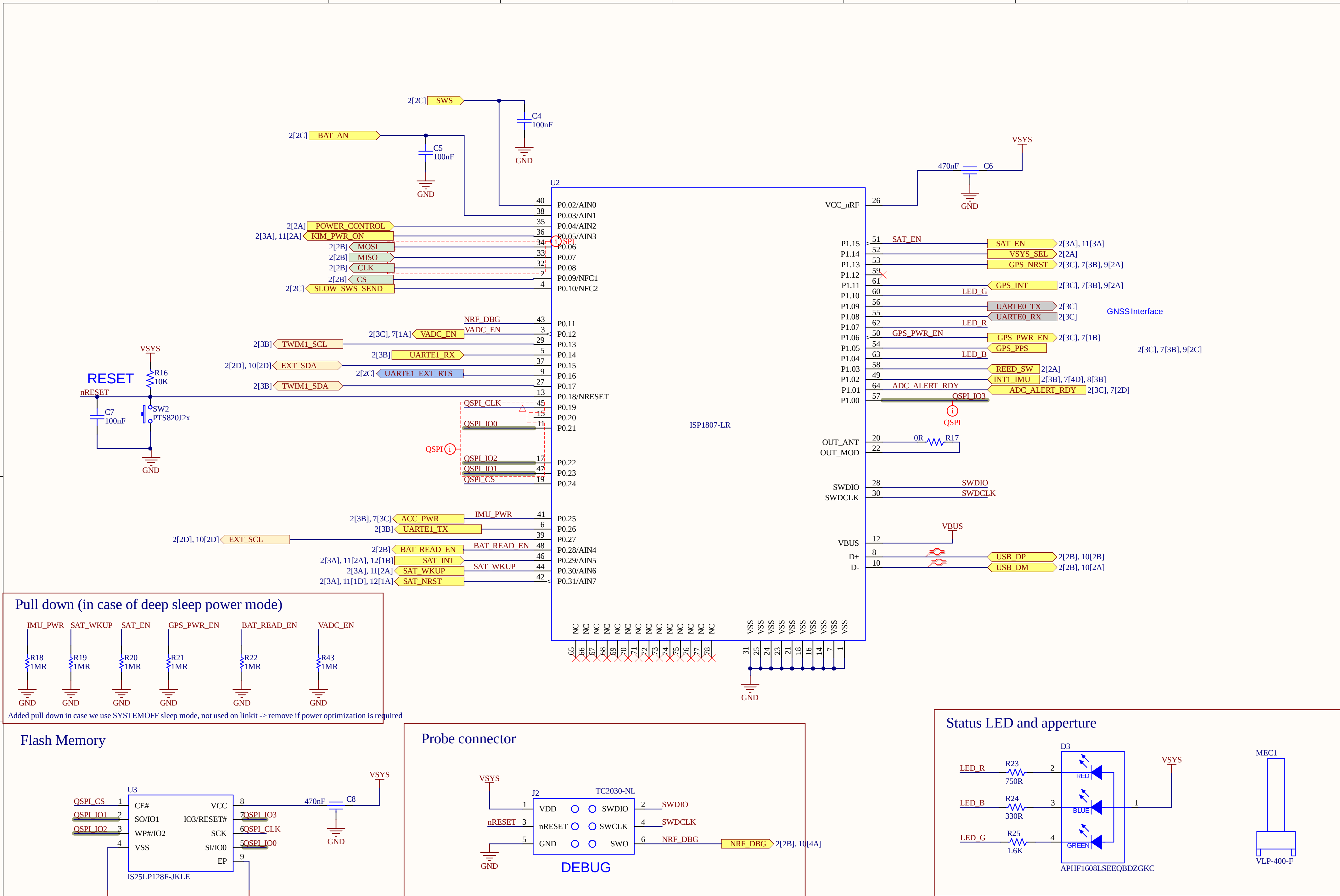

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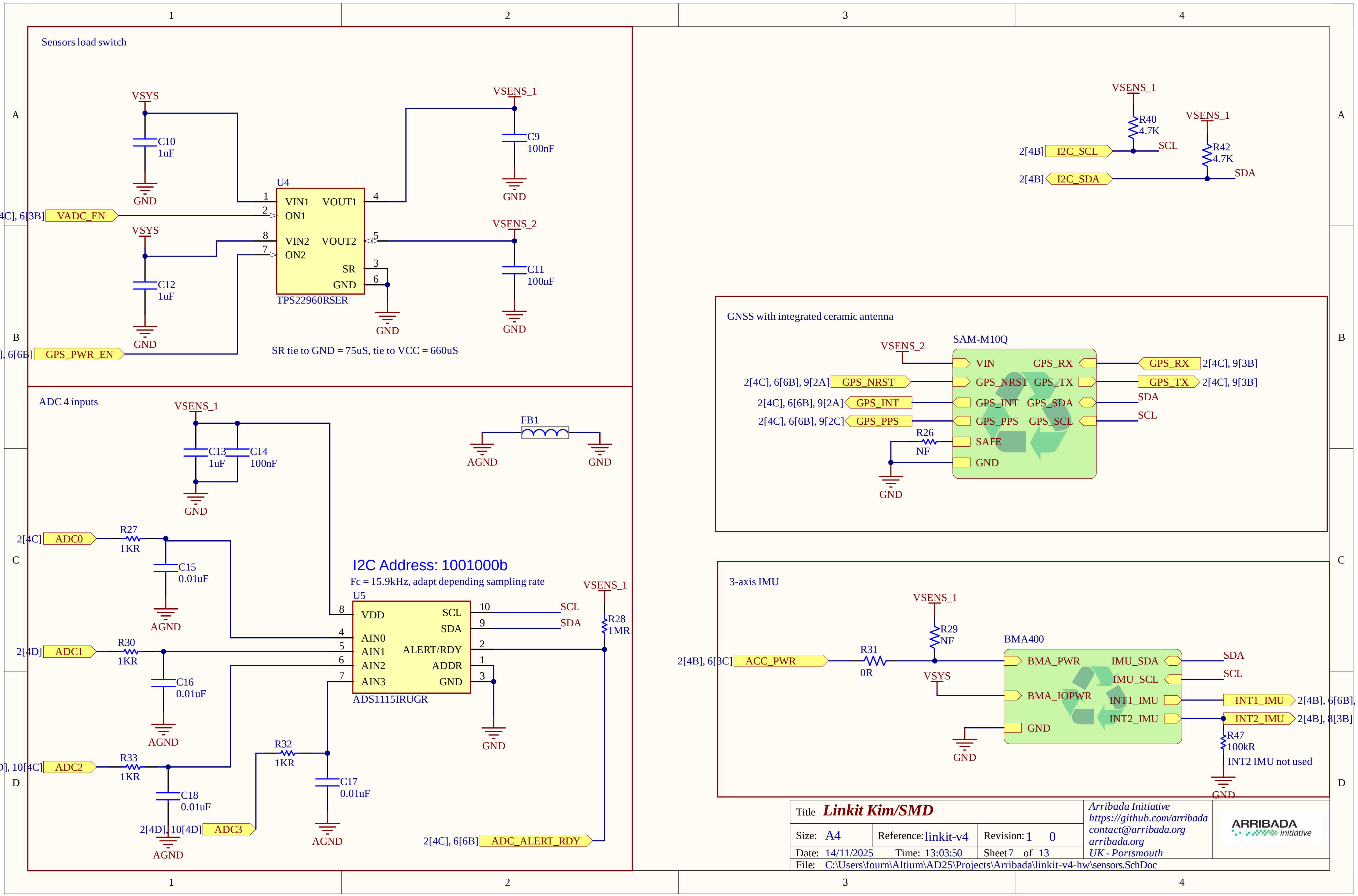
Sensors load switch
VSYS
C10 1uF
GND
VSENS_1
C9 100nF
U4
VIN1 VOUT1
ON1
VIN2 VOUT2
ON2
SR
GND
TPS22960RSER
[4C], 6[3B] VADC_EN
VSENS_2
C11 100nF
C12 1uF
C], 6[6B] GPS_PWR_EN
SR tie to GND = 75uS, tie to VCC = 660uS
ADC 4 inputs
C13 1uF
C14 100nF
FB1
AGND
2[4C] ADC0
R27 1KR
C15 0.01uF
I2C Address: 1001000b
Fc = 15.9kHz, adapt depending sampling rate
U5
VDD SCL
SDA
AIN0
AIN1 ALERT/RDY
AIN2 ADDR
AIN3 GND
ADS1115IRUGR
R28 1MR
2[4D] ADC1
R30 1KR
C16 0.01uF
4D], 10[4C] ADC2
R33 1KR
C18 0.01uF
R32 1KR
C17 0.01uF
2[4D], 10[4D] ADC3
2[4C], 6[6B] ADC_ALERT_RDY
R40 4.7K
R42 4.7K
2[4B] I2C_SCL
2[4B] I2C_SDA
GNSS with integrated ceramic antenna
SAM-M10Q
VIN GPS_RX
2[4C], 6[6B], 9[2A] GPS_NRST GPS_TX
2[4C], 6[6B], 9[2A] GPS_INT GPS_SDA
2[4C], 6[6B], 9[2C] GPS_PPS GPS_SCL
R26 NF SAFE
GND
GPS_RX 2[4C], 9[3B]
GPS_TX 2[4C], 9[3B]
3-axis IMU
BMA400
R29 NF
R31 0R
2[4B], 6[3C] ACC_PWR
BMA_PWR IMU_SDA
IMU_SCL
BMA_IOPWR INT1_IMU
INT2_IMU
INT1_IMU 2[4B], 6[6B], 8
INT2_IMU 2[4B], 8[3B]
R47 100kR
INT2 IMU not used
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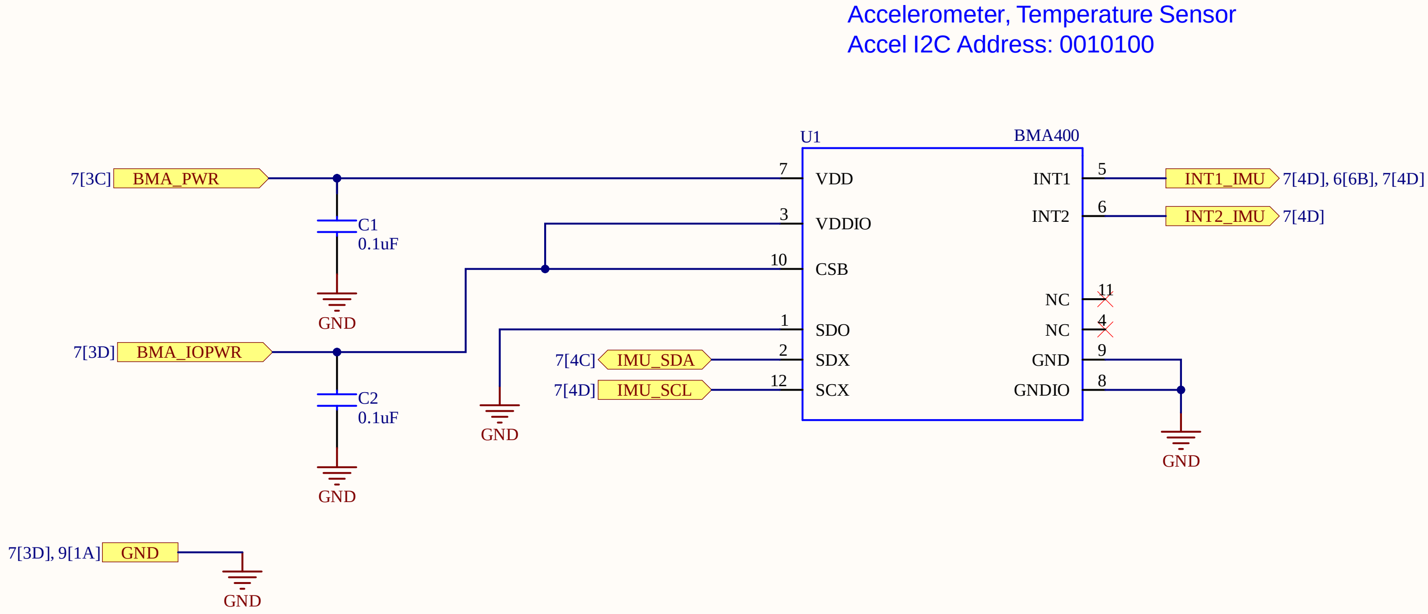

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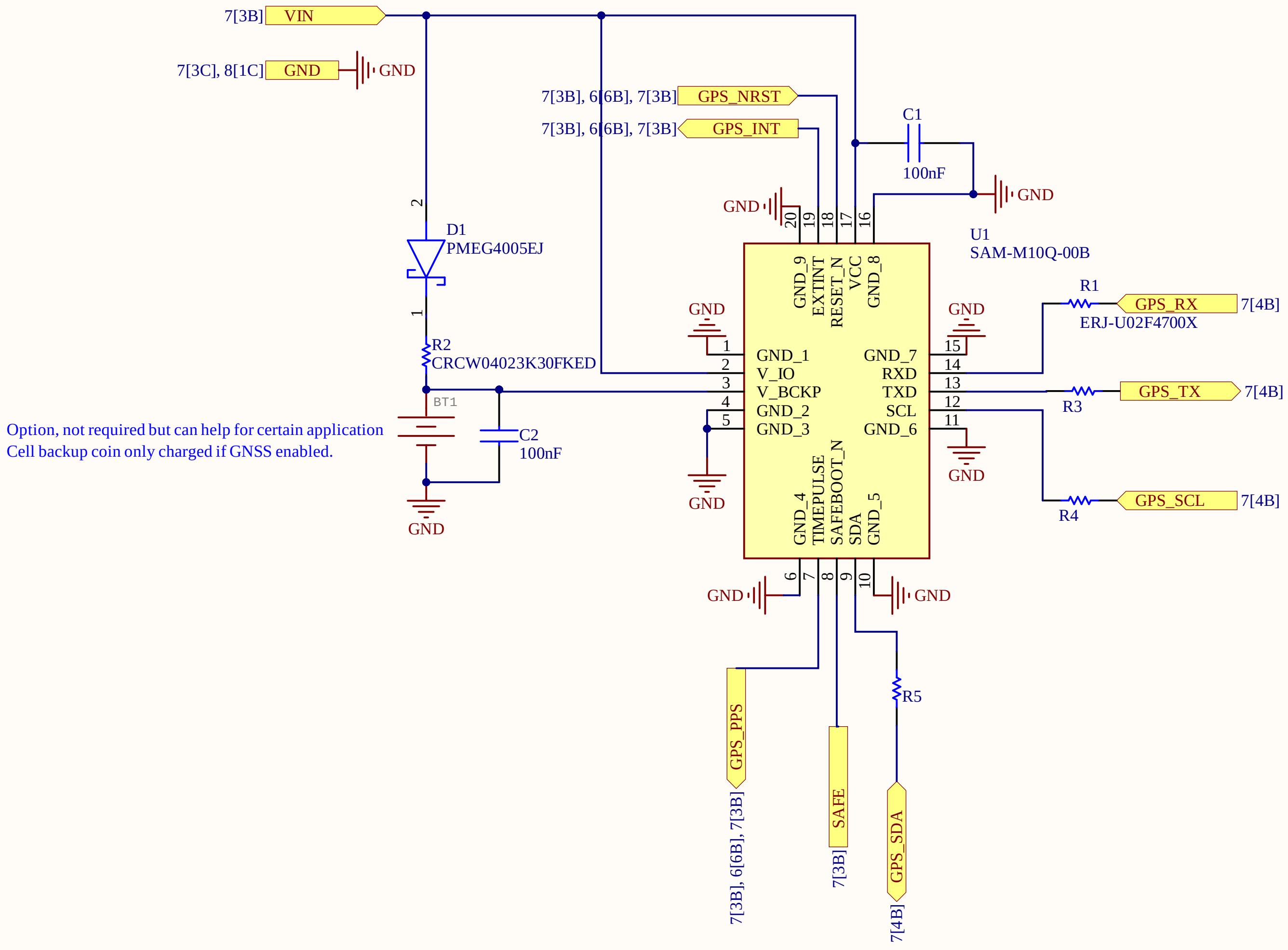

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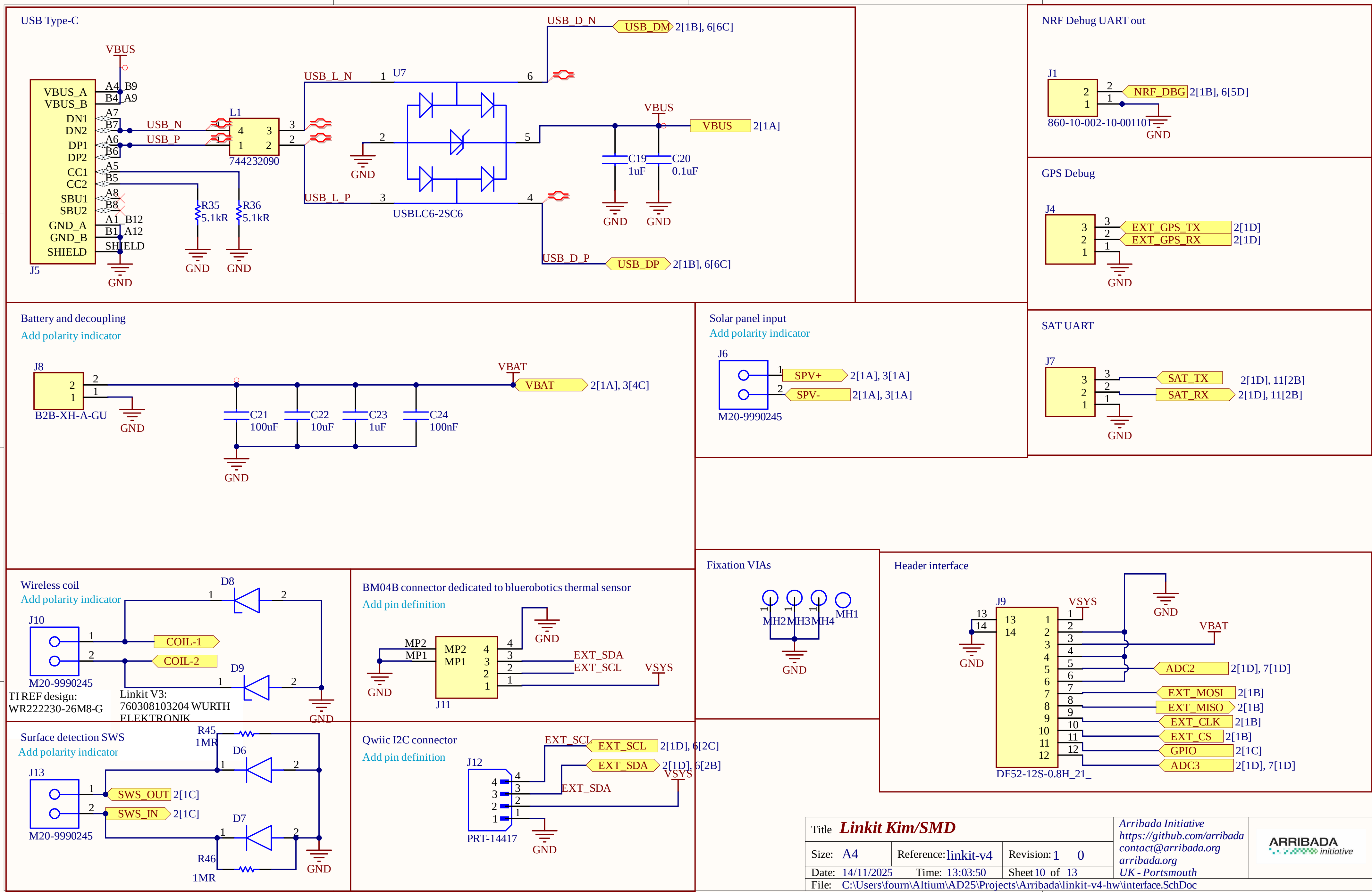

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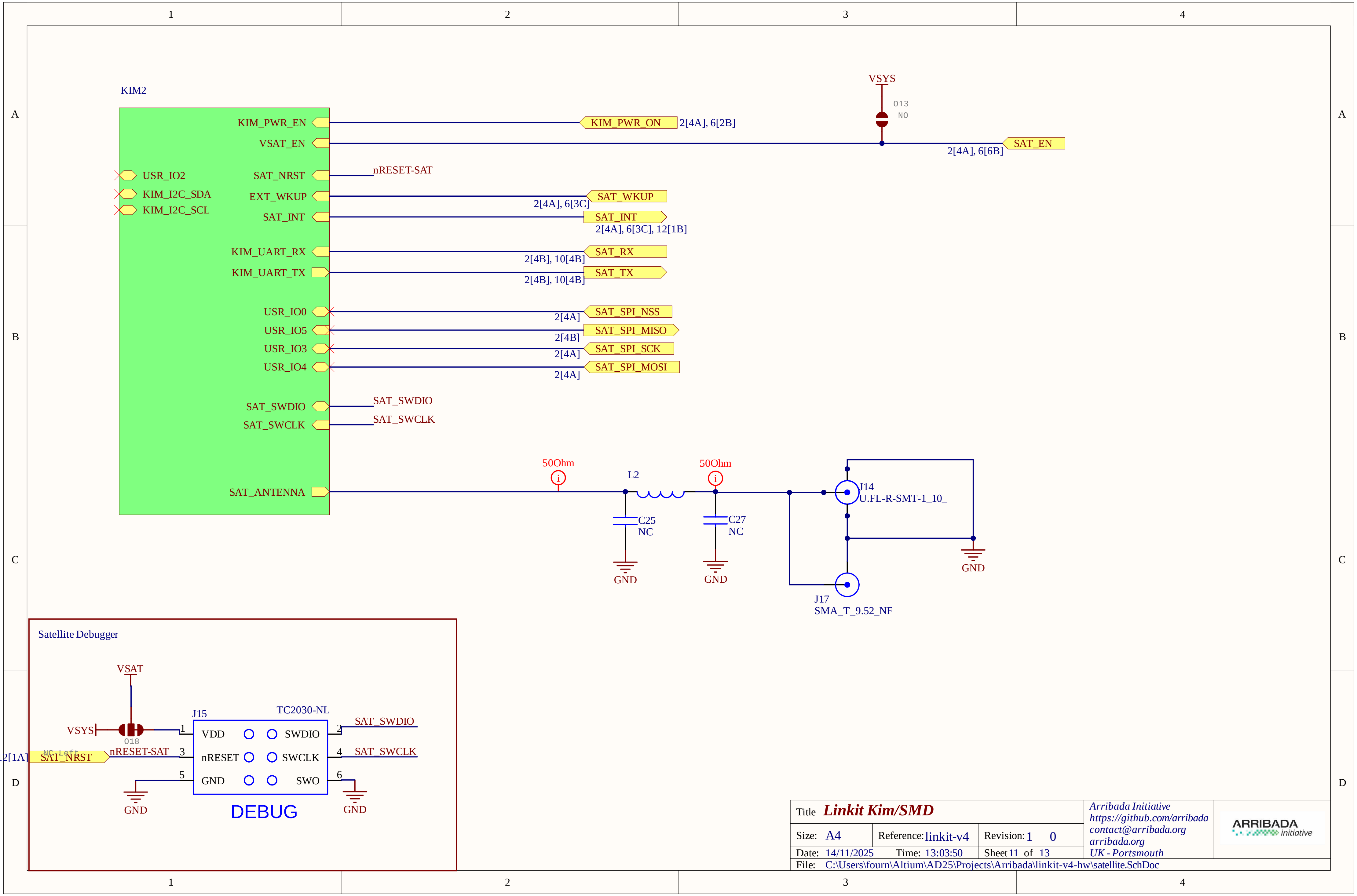
KIM2
KIM_PWR_EN
VSAT_EN
USR_IO2
KIM_I2C_SDA
KIM_I2C_SCL
SAT_NRST
EXT_WKUP
SAT_INT
KIM_UART_RX
KIM_UART_TX
USR_IO0
USR_IO5
USR_IO3
USR_IO4
SAT_SWDIO
SAT_SWCLK
SAT_ANTENNA
KIM_PWR_ON 2[4A], 6[2B]
VSYS
013
NO
SAT_EN
2[4A], 6[6B]
nRESET-SAT
SAT_WKUP
2[4A], 6[3C]
SAT_INT
2[4A], 6[3C], 12[1B]
SAT_RX
2[4B], 10[4B]
SAT_TX
2[4B], 10[4B]
SAT_SPI_NSS
2[4A]
SAT_SPI_MISO
2[4B]
SAT_SPI_SCK
2[4A]
SAT_SPI_MOSI
2[4A]
SAT_SWDIO
SAT_SWCLK
50Ohm
L2
50Ohm
C25
NC
C27
NC
GND
GND
J14
U.FL-R-SMT-1_10_
GND
J17
SMA_T_9.52_NF
Satellite Debugger
VSAT
VSYS
018
J15
TC2030-NL
VDD
SWDIO
nRESET
SWCLK
GND
SWO
SAT_SWDIO
SAT_SWCLK
nRESET-SAT
, 12[1A]
SAT_NRST
GND
GND
DEBUG
Title Linkit Kim/SMD
Size: A4
Reference: linkit-v4
Revision: 1 0
Date: 14/11/2025
Time: 13:03:50
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File: C:\Users\fourn\Altium\AD25\Projects\Arribada\linkit-v4-hw\satellite.SchDoc
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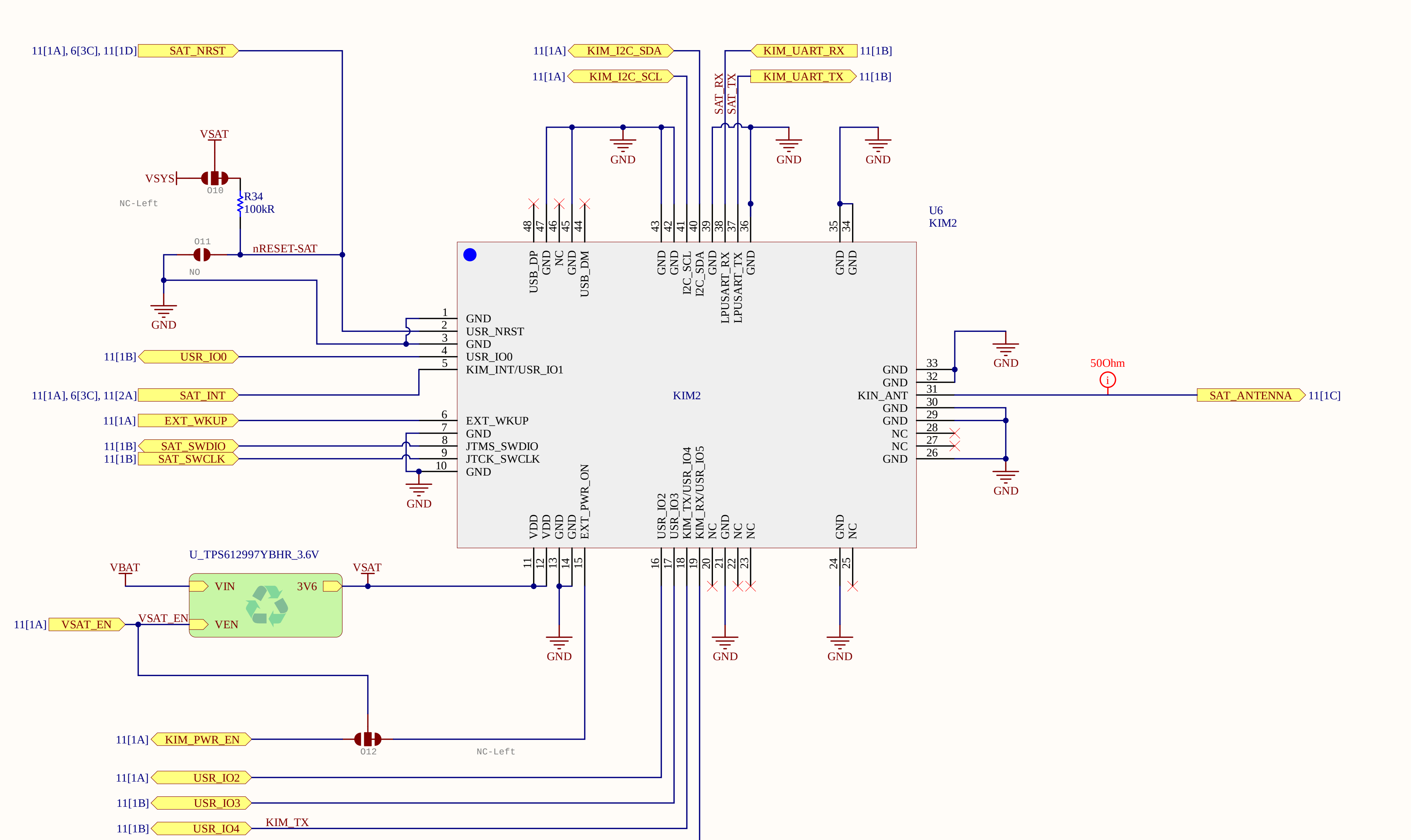
https://github.com/arribada
contact@arribada.org
arribada.org
UK - Portsmouth
ARRIBADA initiative



Title **Linkit Kim/SMD**

Size: A4 Reference: linkit-v4 Revision: 1 0

Date: 14/11/2025 Time: 13:03:50 Sheet 12 of 13

File: C:\Users\fourn\Altium\AD25\Projects\Arribada\linkit-v4-hw\sat-kim2.SchDoc

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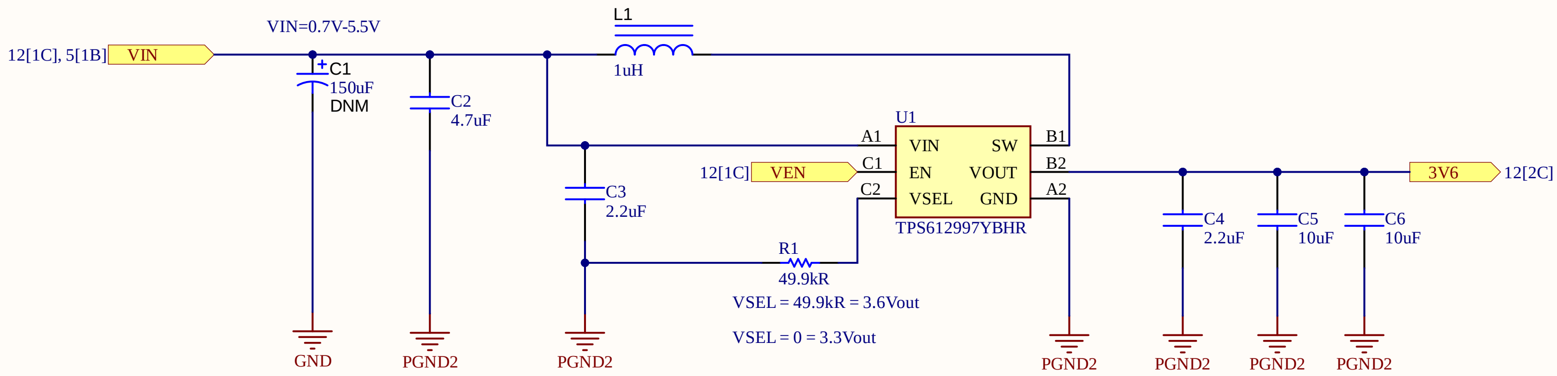

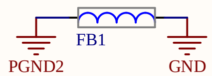

Title **Linkit Kim/SMD**			*Arribada Initiative* *https://github.com/arribada* *contact@arribada.org* *arribada.org* *UK - Portsmouth*
Size: A4	Reference: linkit-v4	Revision: 1 0	
Date: 14/11/2025	Time: 13:03:51	Sheet 13 of 13	
File: C:\Users\fourn\Altium\AD25\Projects\Arribada\hardware-lib\DeviceSheets\Power\TPS612997YBHR_3.6V_lowIQ_1.9A.Sch			

Board Stack Report

Stack Up		Layer Stack			
Layer	Board Layer Stack	Name	Material	Thickness	Constant
1		Top Overlay		0mil	
2		Top Solder	SM-003	1mil	4
3		Top Surface Finish	Nickel, Gold	0.158mil	
4		Top Layer	CF-004	1.378mil	
5		Dielectric 1	S1000-2MB	1.951mil	4.25
6		Dielectric 2	S1000-2MB	1.951mil	4.25
7		Int1 (GND)	CF-004	0.689mil	
8		Dielectric 3	S1000-2MB	1.951mil	4.25
9		Dielectric 4	S1000-2MB	1.951mil	4.25
10		Int2 (Sign)	CF-004	1.378mil	
11		Dielectric 5	Core-001-HCC	41.929mil	4.25
12		Int3 (Sign)	CF-004	1.378mil	
13		Dielectric 6	S1000-2MB	1.951mil	4.25
14		Dielectric 7	S1000-2MB	1.951mil	4.25
15		Int4 (PWR)	CF-004	0.689mil	
16		Dielectric 8	S1000-2MB	1.951mil	4.25
17		Dielectric 9	S1000-2MB	1.951mil	4.25
18		Bottom Layer	CF-004	1.378mil	
19		Bottom Surface Finish	Nickel, Gold	0.158mil	
20		Bottom Solder	SM-003	1mil	4
21		Bottom Overlay		0mil	
	Height : 66.741mil				

Bill of Materials

ARRIBADA initiative

Project Title: Bill of Materials for Variant [linkit-v4-hw-kim] of BOM Document [linkit-v4-hw.BomDoc]
Project File Name: linkit-v4-hw.PrjPcb
Assembly Variant: linkit-v4-hw-kim

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Part Number	Description	Designator	Manufacture	Manufacture Part Number	Supplier	Supplier Part Number	Supplier Unit Price	Quantity
GRM155R61E105	Multilayer Ceramic Capacitors MLCC - SMD/SMT 1 uF 25 VDC 10% 0402 X5R	C1, C21	Murata	GRM155R61E105KE11D				2
GCM188R71C105	Multilayer Ceramic Capacitors MLCC - SMD/SMT 1 uF 16 VDC 10% 0603 X7R AEC-Q200	C2	Murata	GCM188R71C105KA64D				1
CAP CER 100NF 02	CAP CER 100NF 0201 35V 10% X5R	C3, C4, C6	Murata	GRM033R6YA104KE14D	Mouser	81-GRM33R6YA104KE14D		3
CAP FEEDTHRU 47	CAP FEEDTHRU 470NF 0402 6.3V +50%/-20%	C5, C7	TDK	YFF15PC0J474MT000N	Mouser	810-CKD710JB0J474S		2
GRM155R71C104	Multilayer Ceramic Capacitors MLCC - SMD/SMT 0.1 uF 16 VDC 10% 0402 X7R	C8, C10, C13, C22, C27	Murata	GRM155R71C104KA88D	Mouser	81-GRM155R71C104KA88		5
GRM155Z71A105	Multilayer Ceramic Capacitors MLCC - SMD/SMT 1.0UF 10V 10% 0402	C9, C11, C12	Murata	GRM155Z71A105KE01J	Mouser	81-GRM155Z71A105KE1J		3
GRM1555CYA103	Multilayer Ceramic Capacitors MLCC - SMD/SMT 0.01 uF 35 VDC 2% 0402 C0G (NP0)	C14, C15, C16, C57	Murata	GRM1555CYA103GE01D				4
CL03A105KP3NSN	Multilayer Ceramic Capacitors MLCC - SMD/SMT 1uF+/-10% 10V X5R 06 0201	C17	Samsung	CL03A105KP3NSNC				1
GRM033Z71C104	Multilayer Ceramic Capacitors MLCC - SMD/SMT .1UF 16V 10% 0201	C18, C37, C38, C39, C40	Murata	GRM033Z71C104KE14D	Mouser	81-GRM033Z71C104KE4D		5
GRM31CR61A107	Multilayer Ceramic Capacitors MLCC - SMD/SMT 100 uF 10 VDC 20% 1206 X5R	C19	Murata	GRM31CR61A107MEA8L				1
GRM155R61A106	Multilayer Ceramic Capacitors MLCC - SMD/SMT 10 uF 10 VDC 20% 0402 X5R	C20, C54, C55	Murata	GRM155R61A106ME11D				3
GRM188R71E474	Multilayer Ceramic Capacitors MLCC - SMD/SMT 0.47 uF 25 VDC 10% 0603 X7R	C23, C36	Murata	GRM188R71E474KA12D				2
GCJ188R71H563K	Multilayer Ceramic Capacitors MLCC - SMD/SMT 0.056 uF 50 VDC 10% 0603 X7R AEC-Q200	C24	Murata	GCJ188R71H563KA12J				1
GRM155R71E223	Multilayer Ceramic Capacitors MLCC - SMD/SMT 0.022 uF 25 VDC 10% 0402 X7R	C25, C35	Murata	GRM155R71E223KA61D				2
EMK107BBJ106M	Multilayer Ceramic Capacitors MLCC - SMD/SMT 963-MSASE168BB5106MT RPLCMT PN 16V 10uF X5R 0603 20%	C26, C30	Taiyo Yuden	EMK107BBJ106MA-T				2
GRM188R71H103	Multilayer Ceramic Capacitors MLCC - SMD/SMT .01UF 50V 10% 0603	C28	Murata	GCD188R71H103KA01D				1
GRM155R71E103	Multilayer Ceramic Capacitors MLCC - SMD/SMT 0.01 uF 25 VDC 10% 0402 X7R	C29, C34	Murata	GRM155R71E103KA01D				2
GRM188R61C475	Multilayer Ceramic Capacitors MLCC - SMD/SMT 4.7 uF 16 VDC 10% 0603 X5R	C31	Murata	GRM188R61C475KE11J				1
GRM155R71H681	Multilayer Ceramic Capacitors MLCC - SMD/SMT 680 pF 50 VDC 10% 0402 X7R	C32	Murata	GRM155R71H681KA01D				1
GRM155R71H271	Multilayer Ceramic Capacitors MLCC - SMD/SMT 270 pF 50 VDC 10% 0402 X7R	C33	Murata	GRM155R71H271KA01J				1
GRM188R61A106	Multilayer Ceramic Capacitors MLCC - SMD/SMT 10 uF 10 VDC 20% 0603 X5R	C41	Murata	GRM188R61E106KA73D				1
GRM188R61A226	Multilayer Ceramic Capacitors MLCC - SMD/SMT 22 uF 10 VDC 20% 0603 X5R	C42	Murata	GRM188R61A226ME15D				1
C0603C123K5RAC	Multilayer Ceramic Capacitors MLCC - SMD/SMT 50V 0.012uF X7R 0603 10%	C43	KEMET	C0603C123K5RACTU				1
GRM188Z71A475	Multilayer Ceramic Capacitors MLCC - SMD/SMT 4.7 uF 10 VDC 20% 0603 X7R	C51	Murata	GRM188Z71A475ME15D	Digi-Key			1
GRM155R61A225	Multilayer Ceramic Capacitors MLCC - SMD/SMT 2.2 uF 10 VDC 20% 0402 X5R	C52, C53	Murata	GRM155R61A225ME95D				2
GRM155R61H105	Multilayer Ceramic Capacitors MLCC - SMD/SMT 1 uF 50 VDC 20% 0402 X5R	C56, C58	Murata	GRM155R61H105ME05J				2
RB751S40	Schottky Diodes & Rectifiers 40V,Schottky,SOD-523,0.3A	D1	onsemi	RB751S40T1G				1
LED RGB 0603 2M	LED RGB 0603 2MA	D2	Kingbright	APHF1608LSEEQBDZGKC	Mouser	604-APHF1608LQBDZGKC		1
LED RGB 0603 2M	LED RGB 0603 2MA	D3	Kingbright	APHF1608LSEEQBDZGKC	Mouser	604-APHF1608LQBDZGKC		1
PMEG4005EJ	Schottky Diodes & Rectifiers	D4	NXP Semiconductors	PMEG4005EJ115				1
PMEG4010CEAX	Diode	D5, D6	Nexperia	PMEG4010CEAX				2
ESD9L5.0ST5GH	Diode	D7, D8	onsemi	ESD9L5.0ST5G				2
PESD12VS1UL,315	Diode ESD (Uni-directional)	D9, D10	Nexperia	PESD12VS1UL,315				2
BLM18HE601SN1	Ferrite Beads 600 OHM 25%	FB1	Murata	BLM18HE601SN1D				1
BLM18AG102SN1	Ferrite Bead - 1000 OHM 25%	FB2, FB3, FB4, FB5, FB6	Murata	BLM18AG102SN1D				5
BC020-03-A-0200-	Connector	J3	Global Connector Technology	BC020-03-A-0200-0300-L-G				1
USB4215-03-A_RE	USB Connectors USB C Receptacle, Short Body, USB2.0, 16Pin, Horizontal, SMT	J4	Global Connector Technology	USB4215-03-A				1
B2B-XH-A-GU	Connector	J5						1
DF52-12S-0.8H_2	Headers & Wire Housings Recep 12P R/A 0.8mm Pitch	J6	Hirose	DF52-12S-0.8H(21)				1
BM04B-GHS-TBT	Headers & Wire Housings GH HEADER (SMT/TOP TYPE)	J8	JST	BM04B-GHS-TBT(LF)(SN)(N)				1
PRT-14417	QWIIC JST CONNECTOR - SMD 4-PIN	J9	SparkFun	PRT-14417				1
U.FL-R-SMT-1_10	UFL SMT PCB receptable,DC-6GHz Hirose Straight 50 Surface Mount UFL Connector, Receptacle, Solder Termination Coaxial	J11	Hirose	U.FL-R-SMT-1(10)	Mouser	798-U.FL-R-SMT-110		1
744232090	Common Mode Chokes / Filters WE-CNSW 1206 370mA 90Ohm 100MHz AECQ200	L1	Wurth Electronics	744232090				1
DFE252012F-2R2M	Inductor Power Chip Shielded Wirewound 2.2uH 20% 1MHz Metal 2.3A 0.082Ohm DCR 1008 T/R	L2	Murata	DFE252012F-2R2M=P2	Mouser	81-DFE252012F-2R2MP2		1
LQW18AN10NG8	Inductor Wirewound , 10 nH, 52 Milliohm, ± 2%, 4.75 GHz, 1.6 A, 0603 [1608 Metric]	L3	Murata	LQW18AN10NG80D	Digi-Key	490-17899-1-ND		1
XEL4020-102MEC	Power Inductors - SMD 1uH Shld 20% 9.6A 14.6mOhms AECQ2	L5	Coilcraft	XEL4020-102MEB				1
MMBT3904LT1G	Bipolar Transistors - BJT NPN GENERAL PURPOSE	Q1	onsemi	MMBT3904LT1G				1
DMP2305U-7	MOSFET P-CHANNEL ENHANCEMENT MODE	Q2	Infineon	2ED020I12F2XUMA1				1
IRLML6401TRPBF	MOSFETs MOSFT P-Ch -4.3A 50mOhm 10nC Log Lvl	Q3, Q4	Infineon	IRLML6401TRPBF				2
CSD75208W1015	MOSFETs 20V P-channel NexFET Pwr MOSFET A 595-CSD75208W1015T	Q5	Texas Instruments	CSD75208W1015				1
CRCW02011K00F	Thick Film Resistors - SMD 1/20watt 1Kohms 1% 200ppm	R3, R27, R30, R32, R51	Vishay	CRCW02011K00FNED				5
CRCW040210K0J	Thick Film Resistors - SMD 1/16watt 10Kohms 5% Commercial Use	R4	Vishay Semiconductors	CRCW040210K0JNEDC	Mouser	71-CRCW040210K0JNEDC		1
RT0805BRD071M	Thin Film Resistors - SMD 1/8W 1.33M Ohms 0.1%	R5	Yageo	RT0805BRD071M33L				1
RGV3216P-3004-B	Thin Film Resistors - SMD Thin Film Chip Resistors 1206 size, 700V, 0.25W, 3.0M ohm, 0.1%, 25ppm AEC Q200	R6	Susumu	RGV3216P-3004-B-T1				1
RES 10M 0402 1%	RES 10M 0402 1% 100PPM	R7, R9	Yageo Group	RC0402FR-0710ML	Mouser	603-RC0402FR-0710ML		2
CL05A106MP5NU	Multilayer Ceramic Capacitors MLCC - SMD/SMT 10uF+/-20% 10V X5R 1005	R8						1
ERJ-U02F1002X	Thick Film Resistors - SMD 0402 1% 10Kohm Anti-Sulfur AEC-Q200	R10	Panasonic	ERJ-U02F1002X				1
CRCW0201383RF	Thick Film Resistors - SMD 383ohm 1% 200ppm	R11	Vishay	CRCW0201383RFNED				1
CRCW020127K4F	Resistor	R12	Vishay	CRCW020127K4FNED				1
RC0402JR-7D1KL	Thick Film Resistors - SMD 1 kOhms 62.5 mW 0402 5%	R13	Yageo	RC0402JR-7D1KL				1
RC0402JR-7D330	Thick Film Resistors - SMD 330 Ohms 62.5mW 0402 5%	R14	Yageo	RC0402JR-7D330RL				1
CRCW04021K60J	Thick Film Resistors - SMD 1/16watt 1.6Kohms 5%	R15	Vishay Dale	CRCW04021K60JNED				1
RES 10K 0201 1%	RES 10K 0201 1% 100PPM	R16	Vishay Dale	CRCW020110K0FKED	Mouser	71-CRCW020110K0FKED		1
RES 0R 0201	RES 0R 0201	R17	Vishay	CRCW02010000Z0ED	Mouser	71-CRCW02010000Z0ED		1
RC0201FR-7D1ML	Thick Film Resistors - SMD 1 MOhms 50 mW 0201 1%	R18, R20, R21, R22, R28, R57, R59, R60	Yageo	RC0201FR-7D1ML				8
RES 750R 0201 1%	RES 750R 0201 1% 100PPM	R23	Vishay Dale	CRCW0201750RFKED	Mouser	71-CRCW0201750RFKED		1
RES 330R 0201 1%	RES 330R 0201 1% 100PPM	R24	Vishay Dale	CRCW0201330RFKED	Mouser	71-CRCW0201330RFKED		1
RES 1.6K 0201 1%	RES 1.6K 0201 1% 100PPM	R25	Vishay Dale	CRCW02011K60FKED	Mouser	71-CRCW02011K60FKED		1
RES 0R 0402	RES 0R 0402	R31	Yageo Group	RC0402JR-070RL	Mouser	603-RC0402JR-070RL		1
CRCW02015K10J	Thick Film Resistors - SMD 1/20watt 5.1Kohms 5% 200ppm	R33, R34	Vishay Dale	CRCW02015K10JNED				2
CRCW020120K0F	Res Thick Film 0201 453 Ohm 1% 1/20W ±100ppm/°C Molded SMD SMD Paper T/R	R36	Vishay Dale	CRCW020120K0FKED				1
RC0201FR-7D10K	Thick Film Resistors - SMD 10 kOhms 50 mW 0201 1%	R37	Yageo	RC0201FR-7D10KL				1
ERJ-1GNF2550C	Thick Film Resistors - SMD 0201 255ohm 1% HalogenFree AEC-Q200	R38	Panasonic	ERJ-1GNF2550C				1
CRCW0201196RF	Thick Film Resistors - SMD 1/20watt 196ohms 1% 100ppm	R39	Vishay Dale	CRCW0201196RFKED				1
ERJ-U02F4700X	Thick Film Resistors - SMD 1/20watt 470ohms 1% 200ppm	R40, R42	Panasonic	ERJ-U02F4700X				2
CRCW04023K30F	Thick Film Resistors - SMD 1/16watt 3.3Kohms 1% 100ppm	R41	Vishay	CRCW04023K30FKED				1
CRCW020136K5F	Thick Film Resistors - SMD 36.5Kohm 1% 200ppm	R45	Vishay	CRCW020136K5FNED				1
RC0201FR-070RL	Thick Film Resistors - SMD 0 Ohms 50mW 0201 1%	R46	Yageo	RC0201FR-070RL				1
CRCW02012K74F	Thick Film Resistors - SMD 1/20W 2.74Kohms 1% 200ppm	R47	Vishay	CRCW02012K74FNED				1
CRCW0201100KJ	Thick Film Resistors - SMD 1/20watt 100Kohms 5% 200ppm	R48, R63	Vishay Dale	CRCW0201100KJNED				2
CRCW040249K9F	Thick Film Resistors - SMD 1/16watt 49.9Kohms 1%	R49, R50	Vishay	CRCW040249K9FKED				2
CRCW04020000Z	Thick Film Resistors - SMD CRCW 402 jumper	R53, R55	Vishay	CRCW04020000Z0EEHP				2
RES 4.7K 0201 1%	RES 4.7K 0201 1% 100PPM	R54, R56, R61, R62	Vishay Dale	CRCW02014K70FKED	Mouser	71-CRCW02014K70FKED		4
RC0201FR-0710M	Thick Film Resistors - SMD 10 MOhms 50mW 0201 1%	R58	Yageo	RC0201FR-0710ML				1
CT8132BV-IS3	Board Mount Hall Effect/Magnetic Sensors Integrated Digital Omnipolar, TMR, High Sensitivity Magnetic Switch, SOT23 Package, Ind Temp, 2Hz	SW1	Crocus Technology	CT8132BV-IS3				1
PTS820J2x	SW SPST PB 50MA 12V SMT CK variant	SW2	ITT C&K	PTS820J25M5MTRLFS	Mouser	611-PTS820J25M5MTR		1
BQ25176MDSGR	Battery Management 800-mA linear batter y charger for one-ce	U1	Texas Instruments	BQ25176MDSGR				1
ISP1807-LR	IC ISP Smart Bluetooth Module	U2	Insight SiP	ISP1807-LR-ST	Rutronik	ISP1807-LR-ST	$8.50	1
IS25LP128F-JKLE	IC MEM SERIAL NOR FLASH QSPI 128Mb/16MB 3.6V 166MHz WSON8 6X5	U3	ISSI	IS25LP128F-JKLE-TR	Farnell	2901145		1
TPS22960RSER	Power Switch ICs - Power Distribution Low Input VTG Dual L oad Switch	U4	Texas Instruments	TPS22960RSER				1
ADS1115IRUGR	Analog to Digital Converters - ADC 16B ADC w/ Int MUX P GA Comp Osc & Ref A	U5	Texas Instruments	ADS1115IRUGR				1
USBLC6-2SC6	ESD Suppressors / TVS Diodes High Speed Interface Protection Low Cap	U6	STMicroelectronics	USBLC6-2SC6				1
BQ51003YFPT	Wireless Charging ICs Wireless Pwr Receivr A 595-BQ51003YFPR	U7	Texas Instruments	BQ51003YFPT				1
BMA400	IC AXL Accel/temp SPI/I2C 12LGA	U8	Bosch Tools	BMA400	Digi-Key	828-1081-1-ND		1
SAM-M10Q-00B	GNSS / GPS Modules u-blox M10 GNSS antenna module ROM, TCXO, SAW, LNA LGA, 16x16 mm	U9	u-blox	SAM-M10Q-00B				1
TPS63901YCJR	Switching Voltage Regulators 5.5-V buck-boost con verter with 75-nA op	U10	Texas Instruments	TPS63901YCJR				1
KIM2	KIM2 Module 48 pins 1.27mm pitch 36.6 x 25.4 x 5mm	U11						1
TPS612997YBHR	Switching Voltage Regulators 95-nA low IQ 5.5-V b oost converter with INLIMIT (1.9A)	U14	Texas Instruments	TPS612997YBHR				1
							$8.50	151

Approved	Notes